Amrut B Nargund

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ABOUT ME

Final-year Electronics and Communication Engineering student with a minor in Computer Science at PES University. Experienced in mechanical design, robotics development, CAD modelling, and autonomous systems through industrial and research projects. Skilled in Fusion 360, SolidWorks, structural design, and robotics integration, with a strong interest in mechanical product development, automation, and advanced robotic systems.

EXPERIENCE AND PROJECTS

Research Intern – L&T Funded Autonomous Floor Tile Laying Robot

- Designed and developed the complete mechanical architecture of an autonomous industrial floor tile laying robot using Fusion 360 and SolidWorks.
- Engineered the robot chassis, aluminium extrusion frame, payload support structure, and modular assembly for industrial operation.
- Designed tile handling and placement mechanisms incorporating vacuum-based gripping systems for handling heavy floor tiles.
- Performed load calculations, actuator selection, structural analysis, and mechanical component integration to ensure stability and reliability.
- Developed detailed CAD models, manufacturing drawings, and assembly documentation for fabrication and prototyping.
- Collaborated with electronics and software teams to integrate sensors, actuators, and robotic subsystems into the final platform.

UAV Technology Project – IISc UAV Laboratory

- Designed and developed a prototype Unmanned Aerial Vehicle (UAV) platform during training at IISc UAV Laboratory.
- Worked on airframe design, propulsion system selection, weight optimization, and component placement for improved flight stability.
- Applied principles of aerodynamics, flight dynamics, and structural design to develop a functional drone prototype.

Robotics Competition – Bengaluru

- Designed and fabricated mechanical subsystems including drivetrain assemblies, structural components, and mounting mechanisms.
- Implemented sensor integration and hardware-software interfacing for autonomous robot operation.
- Contributed to achieving top 5% performance among participating university teams.
- Applied CAD modelling, rapid prototyping, and mechanical design principles under strict project timelines.

EDUCATION

PES University, Bengaluru, Karnataka 2022 – 2026
B.Tech in Electronics and Communication Engineering (Major) with Minor in Computer Science

NRI Junior College, Guntur 2020 – 2022
Andhra Pradesh Board of Education

SKILLS

Design & CAD: Fusion 360, SolidWorks, Technical Drawings, Mechanical Drafting, GD&T
Mechanical Engineering: Machine Design, Structural Design, Design for Manufacturing (DFM), Design for Assembly (DFA), Material Selection, Tolerance Analysis
Robotics & Automation: ROS, LiDAR Integration, Sensor Integration, Actuator Selection, Pneumatic Systems, Vacuum Handling Systems
Analysis & Engineering: Load Calculations, Mechanism Design, Engineering Calculations
Programming: Python, C, C++, JavaScript
Tools: Git, VS Code, Microsoft Office, Power BI

CORE COMPETENCIES

Mechanical Design — Product Development — CAD Modelling — Robotics Systems
Design Optimization — Problem Solving — Manufacturing Processes — Prototype Development
Team Collaboration — Innovation — Technical Documentation — Engineering Analysis

ADDITIONAL INFORMATION

Hobbies: Automobiles, UAVs and Aerospace Technology, puzzle solving, Robotics, Drawing, Traveling, Financial Analysis, Sports, and Product Design Exploration.